



# NORTH SOUTH UNIVERSITY

DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

## EEE 111L | Analog Electronics I Lab

Fall # 2025

**Course Faculty:** Kazi Safkat Taa Seen [KSE]  
**Lab Instructor:** Sk Raziul Ahasan  
**Email:** raziul.ahasan@northsouth.edu  
**Mobile:** 01558984475 (10 am - 04 pm)  
**Office Room:** SAC 1052 F  
**Office Hours:** W, R 10:00 am – 01:00 pm & 02:30 pm – 04:00 pm  
**Lab Section:** 11  
**Lab Timing:** T 11:20 am – 02:30 pm   **Lab:** SAC504

### Tentative Lab Outline

|       |                                                             |
|-------|-------------------------------------------------------------|
| Lab 1 | I-V Characteristics of Diode                                |
| Lab 2 | Diode Rectifier Circuits                                    |
| Lab 3 | Clipper and Clamper Circuits                                |
| Lab 4 | Zener Diode Application                                     |
| Lab 5 | The Input-Output Characteristics of CE configuration of BJT |
| Lab 6 | The BJT Biasing Circuits                                    |
| Lab 7 | Study of Switching Characteristics of MOSFET                |

### Tentative Marks Distribution

|                                             |                  |
|---------------------------------------------|------------------|
| Class Assessment/Performance                | 5 %              |
| Attendance                                  | 10 %             |
| Lab Report                                  | 20 %             |
| Midterm                                     | 15 %             |
| Final (Written + Hardware Setup Test+ Viva) | (15+30+5) = 50 % |
| Total                                       | 100 %            |

### Tentative Course Schedule:

| Class | Dates     | Experiment                                                                                                                                                                                                                  |
|-------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | 7-Oct-25  | Introductory Class: Grouping, Familiarization with Lab Rules and Safety Procedures and Learn the correct operation and handling of laboratory equipment safely and effectively<br>+<br>Exp 01: I-V Characteristics of Diode |
| 2     | 14-Oct-25 | Exp 02: Zener Diode Application                                                                                                                                                                                             |
| 3     | 21-Oct-25 | Exp 03: Diode Rectifier Circuits                                                                                                                                                                                            |
| 4     | 28-Oct-25 | Exp 04: Clipper and Clamper Circuits                                                                                                                                                                                        |
| 5     | 4-Nov-25  | <b>Midterm</b>                                                                                                                                                                                                              |
| 6     | 11-Nov-25 | Exp 05: The Input-Output Characteristics of CE configuration of BJT                                                                                                                                                         |
| 7     | 18-Nov-25 | Exp 06: The BJT Biasing Circuits                                                                                                                                                                                            |
| 8     | 25-Nov-25 | Exp 07: Study of Switching Characteristics of MOSFET                                                                                                                                                                        |
| 9     | 2-Dec-25  | <b>Practice Session</b>                                                                                                                                                                                                     |
| 10    | 9-Dec-25  | <b>Final</b>                                                                                                                                                                                                                |

**Software:** Multisim 14.3 Professional (x86-x64) Multilanguage [FileCR]